Introduction to Mathematics and Optimization

January, 2021

PROBLEM 1

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be the function given by

$$f(x, y) = x^2 + y.$$

Also define the subset

$$C = \{(x, y) \in \mathbb{R}^2 \mid x \le 1,\ y \le 1,\ x + y \ge 1\}$$

of $\mathbb{R}^2$.

(a) Show that f is a convex function.

(b) Show that f is not strictly convex.

Consider the optimization problem

$$\begin{aligned} &\min f(x, y) \\ &(x, y) \in C. \end{aligned} \tag{1}$$

(c) Is the optimization problem (1) a convex optimization problem? If so, why?

(d) Express the set $U = \mathbb{R}^2 \setminus C$ using strict inequalities (i.e., $>$ and $<$) in x and y. Is U an open subset?

(e) Explain, by a simple reference to the curriculum without calculations, why there must exist an optimal solution to (1).

(f) Solve the optimization problem (1) by **only** using the KKT conditions for (1). Is there more than one solution to the KKT conditions for (1)?

PROBLEM 2

Let $g : \mathbb{R}^2 \to \mathbb{R}$ be the function given by

$$g(x, y) = e^{x^2+y}.$$

(a) Show without using a computer that

$$\frac{\partial^2 g}{\partial x^2} = (2 + 4x^2)e^{x^2+y}.$$

(b) Show that a symmetric 2×2 matrix

$$\begin{pmatrix} a & c \\ c & b \end{pmatrix}$$

with $a, b, c \in \mathbb{R}$ is positive definite if and only if $a > 0$ and $ab - c^2 > 0$.

(c) Show that g is a strictly convex function.

(d) Does the function $g(x, y)$ have a global minimum?

Now let C denote the subset of $\mathbb{R}^2$ given by (as in PROBLEM 1)

$$C = \{(x, y) \in \mathbb{R}^2 \mid x \leq 1,\ y \leq 1,\ x + y \geq 1\}$$

and consider the optimization problem

$$\begin{aligned} &\min g(x, y) \\ &(x, y) \in C. \end{aligned} \tag{2}$$

(e) Why does the optimization problem (2) have a unique solution?

(f) Solve the optimization problem (2).

(g) Does the optimization problem (2) have a unique solution if we replace

$$g(x, y) = e^{x^2+y}$$

with

$$g(x, y) = e^{x+y}?$$

PROBLEM 3

Consider the system of linear equations

$$\begin{array}{ccccccc} x & + & y & + & z & = & 1 \\ x & + & 2y & + & 3z & = & 1 \\ & & y & + & 2z & = & 1 \end{array} \tag{3}$$

in the three variables x, y, and z.

(a) Let

$$v = \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

Find a 3×3 matrix A and a 3×1 matrix b such that the system of equations (3) is expressed as $Av = b$.

(b) Show that there exists no $v \in \mathbb{R}^3$ such that $Av = b$ (the system of equations (3) cannot be solved).

(c) Show that the matrix A is not invertible.

(d) Solve the optimization problem

$$\min_{v \in \mathbb{R}^3} \quad |b - Av|^2.$$

It is permitted to use a computer for matrix calculations here. Is the solution to the optimization problem unique?